THE STATIONARY COVER TIME OF REVERSIBLE CHAINS ON THREE AND FOUR STATES

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Abstract. Aldous and Fill asked whether every reversible Markov chain on n states, started from its stationary law, has expected cover time at least $(n-1)H_{n-1}$, the value for the simple random walk on the complete graph K_n. We prove this for $n = 3$ and $n = 4$, for all reversible chains including those with holding, and show that equality holds only for the simple random walk on K_n. The case $n = 3$ reduces to Schur's inequality. The case $n = 4$ rests on an explicit polynomial identity of degree 16 whose numerator becomes coefficientwise nonnegative after an ordered substitution on each of 30 cones of edge orderings. Holding is handled for every n by an occupation identity: if a chain without holding satisfies the bound and $E_i C \geq (n-1)H_{n-1}/2$ for every state i, then the bound survives any added holding. This pointwise condition holds for $n \leq 4$.

1. Introduction

Let $(X_t)_{t\geq 0}$ be an irreducible reversible Markov chain on a finite set V of n states, with stationary law π. The cover time C is the first time at which every state has been visited; the starting state counts as visited at time 0. Open Problem 6.30 in Aldous and Fill [1] asks whether

$$\mathbb{E}_\pi C \;\geq\; (n-1)H_{n-1}, \qquad H_m = 1 + \tfrac{1}{2} + \cdots + \tfrac{1}{m}, \tag{1}$$

for every reversible chain on n states. The right side is the stationary cover time of the simple random walk on K_n: from a uniform start, the walk waits a geometric number of steps with mean $(n-1)/(n-k)$ for the k-th new state, $k = 1, \ldots, n-1$. The harmonic number enters through Matthews' method [4], which bounds the cover time from any start above and below by H_{n-1} times the largest and smallest expected hitting times between states. For unweighted graphs the asymptotic lower bound of order $n \log n$, from every starting vertex, is due to Feige [3], and Aldous [2] gave lower bounds of the same order for reversible chains started from stationarity; see also [1, Chapter 6]. The exact inequality (1) asks for more, since it compares every weighted chain, with holding allowed, with one explicit extremal chain. For $n = 2$ it is the statement $\mathbb{E}_\pi C \geq 1$, which is clear.

Theorem 1. *Let $n \in \{3, 4\}$. Every irreducible reversible discrete-time Markov chain on n states satisfies $\mathbb{E}_\pi C \geq (n-1)H_{n-1}$, that is $\mathbb{E}_\pi C \geq 3$ for $n = 3$ and $\mathbb{E}_\pi C \geq 11/2$ for $n = 4$. Equality holds if and only if the chain is the simple random walk on K_n: every off-diagonal transition probability is $1/(n-1)$ and there is no holding.*

The proof has three parts. Section 2 writes $\mathbb{E}_\pi C$ as an alternating sum of quadratic forms in inverses of principal submatrices of the weighted Laplacian; this holds for every n. Section 3 shows that holding can be added afterwards, under a pointwise condition that is automatic for $n \leq 4$ (Proposition 3). Holding can lower $\mathbb{E}_\pi C$ for a fixed jump chain (a three-state example is given in Remark 4), so the reduction is not a monotonicity statement. Sections 4 and 5 prove the bound without holding for $n = 3$ and $n = 4$.

2. The stationary cover time as an alternating sum

A reversible chain on V is described by symmetric conductances $w_{xy} = w_{yx} \geq 0$ for $x \neq y$ and holding weights $h_x \geq 0$: with $q_x = \sum_{y \neq x} w_{xy}$ and $m_x = q_x + h_x$, the chain moves from x to $y \neq x$ with probability w_{xy}/m_x and stays at x with probability h_x/m_x. Its stationary law is $\pi_x = m_x / \sum_y m_y$. Every irreducible reversible chain arises this way, with $w_{xy} = \pi_x P_{xy}$ and $h_x = \pi_x P_{xx}$, and irreducibility means that the graph of positive conductances is connected. In this section and in Sections 4 and 5 there is no holding, so $m = q$, and we write $W = \sum_x q_x$.

Let L be the weighted Laplacian, $L_{xx} = q_x$ and $L_{xy} = -w_{xy}$ for $x \neq y$. For a proper nonempty subset $S \subset V$, let L_S be the principal submatrix on S and $q_S = (q_x)_{x \in S}$. If a vector v supported on S has $v^\top L v = \frac{1}{2} \sum_{x,y} w_{xy}(v_x - v_y)^2 = 0$, it is constant on the connected support and vanishes off S, so $v = 0$. Hence L_S is positive definite.

Lemma 2. *For a nonempty set $J \subset V$ with complement $S \neq \emptyset$, let T_J be the hitting time of J. Then $\mathbb{E}_\pi T_J = q_S^\top L_S^{-1} q_S / W$.*

Proof. The function $g_x = \mathbb{E}_x T_J$ vanishes on J and satisfies $g_x = 1 + \sum_y P_{xy} g_y$ for $x \in S$. Multiplying by q_x gives $q_x g_x - \sum_{y \in S} w_{xy} g_y = q_x$, that is $L_S g_S = q_S$. Averaging against $\pi_x = q_x / W$ gives the claim. □

Let T_x be the hitting time of the single state x, with $T_x = 0$ if $X_0 = x$. Then $C = \max_x T_x$. For real numbers t_x, $\max_x t_x = \sum_{\emptyset \neq J \subseteq V} (-1)^{|J|+1} \min_{x \in J} t_x$, and $\min_{x \in J} T_x = T_J$. Taking expectations, using Lemma 2, and writing $S = V \setminus J$ (the term $J = V$ vanishes),

$$W \, \mathbb{E}_\pi C = \sum_{\emptyset \neq S \subsetneq V} (-1)^{n-|S|+1} \, q_S^\top L_S^{-1} q_S. \tag{2}$$

3. Adding holding

Proposition 3. *Let Q be an irreducible reversible chain on n states without holding, with conductances w, and let $c_0 > 0$. Suppose that $\mathbb{E}_\pi C \geq c_0$ for Q and that $\mathbb{E}_x C \geq c_0/2$ for every state x. Then every chain obtained from Q by adding holding weights $h \geq 0$ satisfies $\mathbb{E}_{\pi_h} C_h \geq c_0$. More precisely, with $W_h = W + \sum_x h_x$,*

$$W_h \big(\mathbb{E}_{\pi_h} C_h - c_0 \big) = W \big(\mathbb{E}_\pi C - c_0 \big) + \sum_x h_x \big(2 \, \mathbb{E}_x C - c_0 \big) + h^\top K h \tag{3}$$

for a symmetric matrix K with nonnegative entries.

Proof. For the chain Q let $N_{xy} = \mathbb{E}_x \sum_{t=0}^{C-1} \mathbf{1}\{X_t = y\}$ and $K_{xy} = N_{xy}/q_y$. For each t, the event $\{t < C\}$ depends only on the path $X_0, \ldots, X_t$, through the condition that it misses some state, and this condition is unchanged when the path is reversed. Reversibility gives $q_x \, \mathbb{P}(\text{path}) = q_y \, \mathbb{P}(\text{reversed path})$ for a path from x to y. Summing over paths of length t that miss a state, and then over t, gives $q_x N_{xy} = q_y N_{yx}$, so K is symmetric. Since $\sum_y N_{xy} = \mathbb{E}_x C$, we have $(Kq)_x = \mathbb{E}_x C$.

Add holding weights h and let $m = q + h$. The jump chain of the new chain is Q, and each visit to y before the last new state is reached lasts a geometric number of steps with mean m_y/q_y. The cover time ends on arrival at the last new state, so no waiting is counted there. Conditioning on the jump chain gives $\mathbb{E}_x C_h = \sum_y K_{xy} m_y$, and with $\pi_h(x) = m_x / W_h$,

$$W_h \, \mathbb{E}_{\pi_h} C_h = m^\top K m = q^\top K q + 2 h^\top K q + h^\top K h = W \, \mathbb{E}_\pi C + 2 \sum_x h_x \, \mathbb{E}_x C + h^\top K h.$$

Subtracting $c_0 W_h = c_0 W + c_0 \sum_x h_x$ gives (3). Every term on the right of (3) is nonnegative under the hypotheses. □

Any chain needs at least $n-1$ steps to cover n states, so $\mathbb{E}_x C \geq n-1$. For $c_0 = (n-1)H_{n-1}$ the condition $n-1 \geq c_0/2$ reads $H_{n-1} \leq 2$, which holds for $n \leq 4$ (with $H_3 = 11/6$) and fails for $n \geq 5$. For $n = 3$ and $n = 4$ the middle term of (3) is at least $\sum_x h_x$ and $\frac{1}{2}\sum_x h_x$ respectively, so any positive holding gives a strict inequality. Hence Theorem 1 follows from its case without holding, which is proved in the next two sections.

Remark 4. Adding holding to a fixed jump chain can decrease $\mathbb{E}_\pi C$, because holding also changes the stationary law, and hence the starting distribution. Take three states with $w_{12} = 1$ and $w_{13} = w_{23} = 1/100$, so that state 3 is hard to reach. Without holding $\mathbb{E}_\pi C \approx 100.04$; with holding weight $h_3 = 1$ at state 3 the chain starts there more often, and $\mathbb{E}_\pi C \approx 85.07$. Proposition 3 therefore cannot be replaced by the statement that holding never lowers the stationary cover time.

4. Three states

Write $a = w_{12}$, $b = w_{13}$, $c = w_{23}$, with at most one of them zero. By (2), or by a direct first-step computation,

$$\mathbb{E}_\pi C - 3 = \frac{Q(a,b,c)}{2(a+b+c)(ab+bc+ca)}, \qquad Q = 4(a^3+b^3+c^3) - \sum_{\mathrm{sym}} a^2 b - 6abc, \tag{4}$$

where $\sum_{\mathrm{sym}} a^2 b$ is the sum of the six monomials $a^2b, a^2c, b^2a, b^2c, c^2a, c^2b$. Now

$$Q = \Big(a^3+b^3+c^3+3abc - \sum_{\mathrm{sym}} a^2 b\Big) + 3\big(a^3+b^3+c^3-3abc\big).$$

The first bracket is nonnegative by Schur's inequality and the second by the inequality of arithmetic and geometric means. The second vanishes only when $a = b = c$, and connectivity then forces $a = b = c > 0$. Hence $\mathbb{E}_\pi C \geq 3$ without holding, with equality only for the simple random walk on K_3. Together with Proposition 3 this proves Theorem 1 for $n = 3$.

5. Four states

Label the edges of K_4 as $(x_0, \dots, x_5) = (w_{01}, w_{02}, w_{03}, w_{12}, w_{13}, w_{23})$. For the sets S of size one, $q_S^\top L_S^{-1} q_S = q_x$, and these terms sum to W. For $S = \{i, j\}$,

$$q_S^\top L_S^{-1} q_S = \frac{B_{ij}}{\Delta_{ij}}, \qquad \Delta_{ij} = q_i q_j - w_{ij}^2, \qquad B_{ij} = q_i q_j (q_i + q_j + 2w_{ij}).$$

For $|S| = 3$, the determinant of L_S is the spanning-tree polynomial τ of the weighted K_4, by the matrix-tree theorem. Put $A = \sum_{|S|=3} q_S^\top \operatorname{adj}(L_S)\, q_S$ and $R = \prod_{i<j} \Delta_{ij}$. Then (2) becomes $\mathbb{E}_\pi C = 1 + \big(A/\tau - \sum_{i<j} B_{ij}/\Delta_{ij}\big)/W$, so

$$\mathbb{E}_\pi C - \frac{11}{2} = \frac{F}{2W\tau R}, \qquad F = 2AR - 2\tau \sum_{i<j} B_{ij} \prod_{\{k,l\} \neq \{i,j\}} \Delta_{kl} - 9W\tau R. \tag{5}$$

The denominator is positive: τ and Δ_{ij} are determinants of the positive definite matrices L_S. The numerator F is a homogeneous polynomial of degree 16 in $x_0, \dots, x_5$ with integer coefficients. It has 13,368 terms, of which 2,421 have negative coefficients, so its nonnegativity is not visible term by term.

Proposition 5. *$F(x) \geq 0$ for every $x \in [0,\infty)^6$ whose support is a connected spanning subgraph of K_4, with equality only when all six x_k are equal.*

Proof. Every $x \in [0,\infty)^6$ has some ordering $x_{o_0} \le x_{o_1} \le \cdots \le x_{o_5}$, where o is a permutation of the six edges. Writing $x_{o_k} = z_0 + z_1 + \cdots + z_k$ with $z_k \ge 0$ turns F into a polynomial $G_o(z)$. Relabelling the four states permutes the edges and leaves F unchanged, since it leaves $\mathbb{E}_\pi C$, W, τ and R unchanged. The 720 orderings fall into 30 orbits of size 24 under the 24 relabellings, because no relabelling other than the identity fixes all six edges. It is therefore enough to treat one representative o of each orbit.

For each of the 30 representatives, the polynomial G_o has nonnegative integer coefficients; there are 577,620 nonzero coefficients in all, recorded in the accompanying files. This proves $F \ge 0$. For equality, note that in every G_o the monomials $z_0^{14} z_j^2$, $1 \le j \le 5$, have positive coefficients. If all edges are positive then $z_0 > 0$, so $F = 0$ forces $z_1 = \cdots = z_5 = 0$, which means all six weights are equal. If some edge vanishes, let $j \ge 1$ be the smallest index with $x_{o_j} > 0$; then $z_j > 0$, and the positive edges are exactly $o_j, \dots, o_5$. Whenever these edges form a connected spanning subgraph, the coefficient of z_j^{16} in G_o is positive, so $F > 0$. □

By (5) and Proposition 5, $\mathbb{E}_\pi C \ge 11/2$ for every irreducible reversible chain on four states without holding, with equality only when all conductances are equal, that is for the simple random walk on K_4. With Proposition 3 this proves Theorem 1 for $n = 4$.

Remark 6. Proposition 5 is a finite computation. The coefficients of the 30 polynomials G_o, the polynomial F, and programs that rebuild F from (2) and check every coefficient in exact integer arithmetic are in the repository listed below. Two further checks are independent of the program that produced the certificate: a second reconstruction of F from integer dictionaries, Leibniz determinants and an enumeration of spanning trees, and an evaluation of each G_o at random integer points against $2W\tau R(\mathbb{E}_\pi C - 11/2)$ computed by an exact solve of the first-step equations on pairs (current state, visited set). Exact enumeration over all 624 connected weightings in $\{0,1,2\}^6$, each with three holding vectors, finds minimum $11/2$, attained only at the two constant weightings without holding. Lean 4 with Mathlib checks the three-state cover-time formula for strictly positive conductances and nonnegative holding, and the implication about symmetric matrices used in Proposition 3. The occupation identity, the cases with a zero conductance and the case of equality are proved above.

6. Remarks on larger n

For $n \ge 5$ the trivial bound $\mathbb{E}_x C \ge n - 1$ falls short of $(n-1)H_{n-1}/2$, so the hypothesis of Proposition 3 would have to be proved separately, and the reduction from chains with holding is open. The approach of Section 5 also grows quickly: for $n = 5$ there are ten edges and $10!/120 = 30{,}240$ orbits of edge orderings. Numerical minimisation over reversible chains on five and six states, with holding, has found no value below $(n-1)H_{n-1}$.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [7] and Math Brain [5]. In October and November 2026 Claude agents (Anthropic) and Codex agents running OpenAI's GPT-6.1, connected to the Brain and working on the same problem, studied the stationary cover time of small reversible chains. A Claude agent proved the case $n = 3$; a Codex agent found the formula (5), the cone certificate for $n = 4$ and the occupation identity of Proposition 3. Claude agents checked the certificate with independent programs, formalised in Lean the case $n = 3$ with positive conductances and the matrix statement behind the holding lemma, and Claude Opus 5.5 (Anthropic) prepared the manuscript and audited it. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code, certificates and Lean files are at
`https://github.com/yspennstate/stationary-cover-time-four-states`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [5].

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